

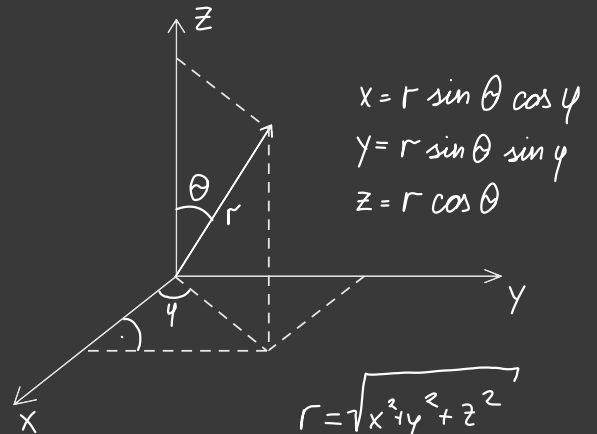
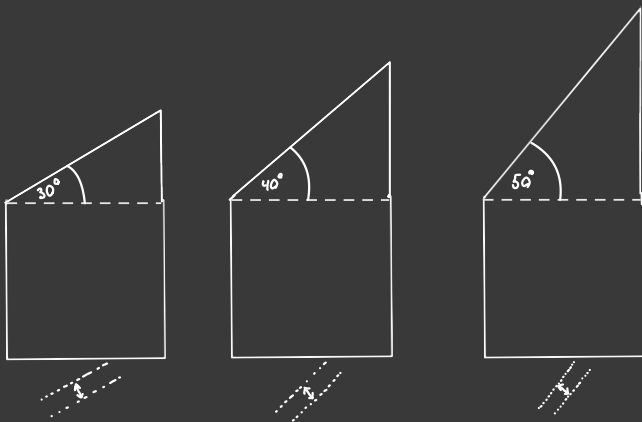
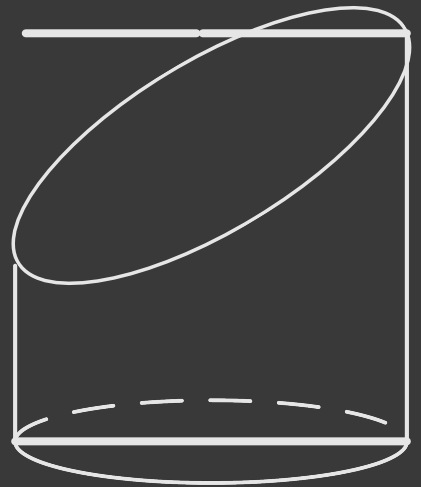
NEMA 17 17HS8401

Step Angle 1.8°

Step Angle Accuracy: $0.09^\circ (5\%)$

$$\frac{0.09^\circ}{360^\circ} \cdot 2\pi \cdot 12\text{m} = 0.0188\text{m} = 1.9\text{cm error}$$

$$\frac{1.8^\circ}{360^\circ} \cdot 2\pi \cdot 12\text{m} = 0.3768\text{m} = 37.7\text{cm resolution at } 90^\circ$$

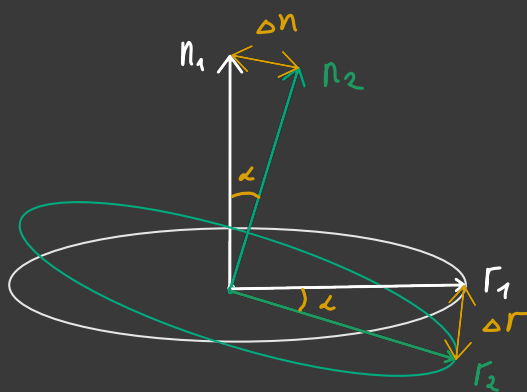


$$\begin{aligned} x &= r \sin \theta \cos \varphi \\ y &= r \sin \theta \sin \varphi \\ z &= r \cos \theta \end{aligned}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \arctg\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$$

$$\varphi = \arctg\left(\frac{y}{x}\right)$$



fixed $\Delta\varphi/\text{step}$ for servo (1.8°)

$$\angle = \angle(\theta, \Delta\varphi)$$

$$\Delta r = \Delta r(r, \theta, \Delta\varphi)$$

$$0.) \quad n_0 = (x_0, y_0, z_0) \quad |n_0| = 1$$

$$x_0 = \sin \theta \cos \varphi$$

$$y_0 = \sin \theta \sin \varphi$$

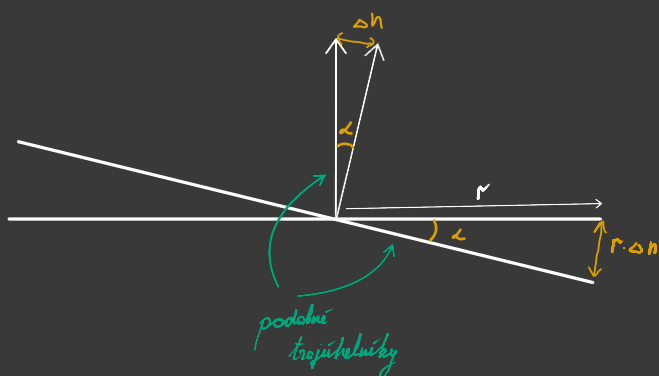
$$z_0 = \cos \theta$$

$$1.) \quad n_1 = (x_1, y_1, z_1) \quad |n_1| = 1$$

$$x_1 = \sin \theta \cos(\varphi + \text{step})$$

$$y_1 = \sin \theta \sin(\varphi + \text{step})$$

$$z_1 = \cos \theta$$



$$|\Delta n| = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2} = \sin \theta \sqrt{(\cos(\varphi + \text{step}) - \cos \varphi)^2 + (\sin(\varphi + \text{step}) - \sin \varphi)^2}$$